CHAPTER 10

Handling Errors From Increased Concurrency

In this chapter you'll see some common challenges faced with higher levels of concurrent use of the database. More concurrency comes with higher query volume. Your database will begin to have many more client connections and this will make it more critical that queries run quickly. You will also likely to see more errors due to lock contention or queries that are canceled, including cancellations due to safeguards you add.

Managing database connections is an important part of managing your server resources. Database connections are a limited resources and as the query volume increases you may begin to use all of the available connections and even need more connections than what's available.

You'll see how to do that in this chapter. You'll add safeguards that help prevent bad queries from consuming too many resources. You'll see how to better use the limited set of database connections that are available.

Although you may not have these issues today, by developing awareness and practicing how to deal with them in advance, you'll be well positioned to solve them if they arise.

Concurrency Terminology



- Connection Pooler — Middle layer software between Rails and PostgreSQL, to more efficiently use database connections
- Server Connections — Database connections
- Client Connections — Active Record pool client connections